00164

Sri Lanka Institute of Information Technology

B.Sc. Honours Degree in Information Technology

Specialized in Information Technology

Final Examination
Year 3, Semester 2 (2022)

IT3080 – Data Science and Analytics

Duration: 2 Hours

November 2022

Instructions to Candidates:

- This paper is preceded by 10 minutes reading period. The supervisor will indicate when answering may commence.
- This paper has 4 questions.
- Answer all questions in the booklet given.
- The total marks for the paper is 100.
- This paper contains 5 pages, including the cover page.
- Calculators are allowed. Any other electronic devices capable of storing and retrieving text, including mobile phones are not allowed.

Question 1 — 30 Marks

a. What is meant by data-driven decision marking and how is it helpful for decision makers? (2 marks)

b. What is meant by missing completely at random (MCAR) in missing data? What are the suitable techniques to handle missing data when it is of MCAR type? (3 marks)

c. A dataset contains 500 attributes and 100 rows of data. What kind of issues may result during data analysis with this dataset? Briefly describe the actions that could be taken by the data analyst to resolve such issues. (4 marks)

d. State the 'A Priori' property in Association Rule Mining. What is it useful for? (2 marks)

e. Consider the following transactions at a store.

1	Bed Spread, Towel, Pillow, Mosquito Net
2	Pillow, Pillowcase, Carpet
3	Mattress, Towel, Bed Spread, Carpet
4	Mosquito Net, Tablecloth, Towel, Bed Spread
5	Tablecloth, Oven Mitts, Pillow
6	Pillow, Pillowcase, Carpet, Towel
7	Oven Mitts, Towel, Bed Spread, Tablecloth
8	Tablecloth, Bed Spread, Pillow, Pillowcase

i. Assuming the minimum support is 40%, identify the frequent item sets within the set of transactions following the A priori algorithm. (8 marks)

ii. Assuming that minimum confidence is 40%, determine whether Bedspread → pillow is a strong association rule. (3 marks)

f. Briefly explain the Agglomerative hierarchical clustering method. (2 marks)

g. Consider the following 4 observations in a 2D space. The distance between observations calculated using Euclidean distance is given.

	A	B	C	D
A	0	15	2	3
B	15	0	5	8
C	2	5	0	6
D	3	8	6	0

Use single link agglomerative clustering to group the data. Show how the clustering is performed step by step. Show the dendrograms also in your answer. (6 marks)

Question 2 (30 marks)

a. Briefly explain Supervised and Unsupervised learning by proving examples. (3 marks)

b. Consider the following dataset which is used to predict whether a customer would deposit in a given type of account based on his/her job, marital status, and level of education.

[illegible]	Marital status	[illegible]	Deposit
Managerial	married	secondary	No
Managerial	Single	secondary	No
Executive	Single	secondary	Yes
Executive	married	secondary	No
Managerial	married	tertiary	Yes
Executive	single	tertiary	Yes
Executive	married	tertiary	Yes
Managerial	Single	secondary	No

iii. Construct a binary decision tree to do the above prediction based on the given dataset showing calculations required for adding branches to the tree. (14 marks)

iv. Based on the decision tree construct, would a customer with secondary education in a managerial position deposit in the account? (2 marks)

c. What is meant by the seasonal component in a time series? Provide an example where a seasonal component may be seen in a time series. (2 marks)

d. Consider the following outputs obtained by executing three models used to predict profit in an organization using regression techniques.

Model 1

Dep. Variable:	Profit	R-squared:	0.947
Model:	OLS	Adj. R-squared:	0.945
Method:	Least Squares	F-statistic:	849.8
Date:	Tue, 18 Oct 2022	Prob (F-statistic):	3.50e-32
Time:	11:47:36	Log-Likelihood:	-527.44
No. Observations:	50	AIC:	1059.
Df Residuals:	48	BIC:	1063.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	4.903e+04	2537.897	19.320	0.000	4.39e+04	5.41e+04
R&D Spend	0.8543	0.029	29.151	0.000	0.795	0.913

Model 2

Dep. Variable:	Profit	R-squared:	0.040
Model:	OLS	Adj. R-squared:	0.020
Method:	Least Squares	F-statistic:	2.015
Date:	Tue, 18 Oct 2022	Prob (F-statistic):	0.162
Time:	11:49:26	Log-Likelihood:	-599.63
No. Observations:	50	AIC:	1203.
Df Residuals:	48	BIC:	1207.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	7.697e+04	2.53e+04	3.040	0.004	2.61e+04	1.28e+05
Administration	0.2887	0.203	1.419	0.162	-0.120	0.698

Model 3

Dep. Variable:	Profit	R-squared:	0.948
Model:	OLS	Adj. R-squared:	0.946
Method:	Least Squares	F-statistic:	426.8
Date:	Tue, 18 Oct 2022	Prob (F-statistic):	7.29e-31
Time:	11:50:58	Log-Likelihood:	-526.83
No. Observations:	50	AIC:	1060.
Df Residuals:	47	BIC:	1065.
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	5.489e+04	6016.718	9.122	0.000	4.28e+04	6.7e+04
R&D Spend	0.8621	0.030	28.589	0.000	0.801	0.923
Administration	-0.0530	0.049	-1.073	0.289	-0.152	0.046

i. Based on the above outputs what is the best model which could be used to predict the profit in the organization? Explain how you made the conclusion. (5 marks)

ii. How much of the variance of Profit is explained by the best model? (1 mark)

iii. Assume that the amount of money spent is given in Millions, what would be the profit if the Money spent on Administration is 35 million and the money spent on R & D is 10 Million? (3 marks)

Question 3 (15 Marks)

a. List 3 advantages of using Hadoop? (3 marks)

b. Briefly explain the re-replication in HDFS using an example. (4 marks)

c. You have just been hired by YouTube to help with analyzing their large amounts of log data. YouTube produces 4 text files with log data. Each line of one of these files contains video ID, category, length, and views.

The following input file contains a sample of data that can be used for the analysis.

Input

1	Music	15	12k
2	Food	40	9k
3	Music	20	13 k
4	Travel	60	20 k
5	Sports	35	5 k
6	Music	12	10k

i. Draw MapReduce workflow to calculate the most-watched category in the given data set. Indicate the results in split, map, combine, shuffle/ sort, and reduce phases. Assume that this involves two mappers. (5 marks)

ii. What is the <key, value> in the map phase? (1 mark)

iii. What is the <key, value> in the combine phase? (1 mark)

iv. What is the <key, value> in reduce phase? (1 mark)

Question 4 (25 Marks)

a. Briefly explain a difference between a database and a Data warehouse explaining why the differences exist in each of them. (2 marks)

b. What are the two major elements in a Dimensional Model? Explain the purpose of each element briefly. (2 marks)

c. Briefly explain the elements of a star schema and its arrangement of elements and connection between elements. (3 marks)

d. Consider a large private university in Europe that enrolls a large number of students from across the. Think about what data would be of interest to the university to analyze its business potential. The university hopes to store its historical data in a data warehouse in order to do the above.

i. Draw a logical design for the above data warehouse. The schema should contain at least four-dimension tables. The fact table should contain at least 3 metrics and the dimension tables should contain at least four attributes each. (14 marks)

ii. Assume that a multidimensional model is developed on top of the data warehouse, explain an operation which would be useful for the university using an illustration. (4 marks)

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